

Retrieval quality note

Re-tested 2026-07-20 against the live Azure AI Search index (19 docs, 481 chunks, all short-named per `design-doc.md` §2's naming audit) via the full `POST /api/v1/chat` endpoint — this table shows what the user actually sees: the final, per-citation-filtered sources alongside the answer. Score = Azure's semantic reranker score (range ~0-4, not a 0-1 similarity). Covers all 4 corpus formats and out-of-corpus/negative cases.

#	Question	Cited (filtered to what the answer actually used)	Score	Notes
1	How many days per week can employees work remotely?	(none)	—	Answer opens with "I don't have that information" — none of the sources specify a fixed day count, so citations are correctly suppressed even though real handbook sources were retrieved (see "Tuning applied").
2	What is the progressive discipline policy?	<code>progressive-discipline-policy</code> , "Policy brief & purpose"; <code>progressive-discipline-policy</code> , "How to invoke progressive discipline"	3.053, 2.814	Exact match; confirms the mislabeled-extension file (HTML content, no <code>.html</code> extension) is retrievable, not just parseable.
3	What counts as an excused absence under the attendance policy?	(none)	—	Answer opens with "I don't have that information" — the attendance template doesn't enumerate excused-absence examples.
4	What are examples of prohibited or risky uses of AI at work?	<code>ai-policy.pdf</code> , Page 3; <code>ai-policy.pdf</code> , Page 2	2.889, 2.402	Correct.
5	What is required when requesting a remote work arrangement?	<code>hr-manual.docx</code> , "Flexible Working Arrangements"; <code>remote-work.docx</code> , "Eligibility"	2.666, 2.359	Correct; validates DOCX heading-based chunking (<code>Eligibility</code> came from a Word "Heading" style, not raw text).
6	How do I request FMLA leave and what medical certification is required?	<code>handbook-nonprofit.pdf</code> , Page 24; <code>fmla-guide.pdf</code> , Pages 15/14/12/10	3.017, 2.994- 2.85	Correct; <code>fmla-guide.pdf</code> is the DOL's dedicated FMLA guide (formerly <code>employeeeguide.pdf</code> — its old name was generic and misleading).
7	What onboarding	<code>onboarding-checklist.html</code> ,	2.57, 2.271,	Correct; three different formats (HTML, Markdown, PDF) all surface relevant onboarding

#	Question	Cited (filtered to what the answer actually used)	Score	Notes
	steps happen for a new hire?	"Onboarding checklists & templates"; <code>career-growth.md</code> , "Your First Year"; <code>handbook-howto-gusto.pdf</code> , Page 13	2.157	content.
8	What career growth or advancement opportunities are described for employees?	<code>career-growth.md</code> , "Mastery & Titles"	2.272	Correct; the dedicated career-growth Markdown doc is the top hit.
9	What are the benefits of a retired employee?	(none)	—	The original bug report, re-tested: previously this question surfaced a <code>wfahandbook.pdf</code> chunk defining <i>federal-employee</i> separation/retirement codes — plausible-looking but from a document that had no business being in this corpus. That document is gone; the answer now honestly opens with "I don't have that information" for retiree-specific benefits, so citations are suppressed even though real 401(k)/benefits sources were retrieved underneath.
10	How do policy changes get proposed? → (follow-up) "tell me more about that"	<code>policy-changes.md</code> , "Proposals"/"Discussion Meetings"/"Continuing Work" (turn 1); same + "Slack #handbook" (turn 2)	2.489-2.361 → 2.534-1.847	Multi-turn follow-up correctly resolved via <code>generation_service.condense_query()</code> .
11	What is the capital of France?	(none — out of corpus)	—	Correctly refuses.
12	What is our company's stock ticker symbol?	(none — plausible-sounding, out of corpus)	—	Correctly refuses despite sounding like a legitimate HR-adjacent question.

Tuning applied

- **Corpus naming, round 2 (2026-07-20):** with 9 of 19 files touching "benefits" or "handbook" in some way (e.g. `employee-benefits-summary-template.docx` , `shrm-sample-employee-handbook-2023.docx`), the descriptive-but-long names from round 1 were still hard to visually tell apart. Shortened everything with a consistent `benefits-` / `handbook-` prefix scheme — see `design-doc.md` §2 for the full mapping. Re-ingested after renaming since checksums (and therefore `upsert_document` 's dedup key) are unchanged by a rename — the database and search index needed a full rebuild to pick up the new filenames at all, not just an incremental re-index.
- **Corpus audit, round 1 (2026-07-20):** every document was opened and read, not trusted by filename. Found `deo_handbook.pdf` and `wfahandbook.pdf` were federal-agency internal-operations manuals (hiring-examiner procedures; workforce-analysis methodology) with nothing to do with employee HR policy, making up 58% of the entire chunk index — row #9 above is a direct before/after of the accuracy problem this caused. Both removed; three more files renamed to match their real content.
- **Citation display, "leads with disclaimer":** rows #1, #3, and #9 all retrieve real, on-topic sources, but the model's answer opens with "I don't have that information" because none of them specify the exact thing asked (a day count, specific excused-absence examples, retiree-specific benefits). Showing a Sources panel on an answer whose headline is a disclaimer read as contradictory, so citations are now dropped whenever the answer's opening sentence is that kind of disclaimer — not just on a full refusal. See `guardrail_service.leads_with_disclaimer()` .
- **Citation filtering, bracket bundling:** the model doesn't always cite one bracket per source — it sometimes bundles several as `[fileA, refA; fileB, refB]` . Citations are matched by checking each bracket group for the filename and chunk_ref co-occurring, not by requiring an exact one-bracket-per-citation string match.
- **Generation speed/verbosity:** GPT-5 defaulted to a heavy reasoning effort with no `reasoning.effort` / `text.verbosity` hint — measured 28+ seconds of silent "thinking" before the first token on one turn. Tuned to `reasoning.effort="minimal"` , `text.verbosity="low"` for GPT-5 (GPT-5.5 uses `"none"` / `"low"` ; DeepSeek V3.2 accepts neither param). Cut first-token latency from ~28-30s to ~3-6s across all three models, verified live against each.
- **Refusal threshold:** 2.0 on Azure's semantic reranker scale (~0-4), separating in-corpus scores (2.2-3.1 in this test set) from out-of-corpus scores (below 2.0, rows #11-12).
- **Content-type sniffing:** `progressive-discipline-policy` and `handbook-sample.doc` have non-HTML extensions but are actually HTML — the loader sniffs actual content type rather than trusting the extension (row #2 confirms this works at retrieval time, not just parse time). Kept under these exact names deliberately as a live test of that behavior.
- **Chunking:** DOCX heading styles (`Heading 1` / `Heading 2`) are preserved as section titles in `chunk_ref` (row #5's "Eligibility"), giving citations a human-readable section name instead of a raw chunk index.